

Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	
Project Name	AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform indentation and formatting
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend and backend are organized
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve code readability and maintenance.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named based on their s
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added for important functions and complex logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into reusable React components, FastAPI routes, and service modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unused code has been removed to improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented for reliable execution.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	Authentication Module	FastAPI, React.js	Login & Registration	High
2	Loan Management Module	FastAPI, React.js	Dashboard, Loan Management	High
3	Financial Health Engine	Python (FastAPI)	Dashboard, Financial Health Analysis	High
4	AI Integration Module	Google Gemini AI, Python	Settlement Recommendations, AI Negotiation, AI History	High
5	UI Components	React.js	Navigation Bar, Forms, Dashboard Cards, Buttons	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend, backend, and database modules.
Readability	5	Meaningful naming conventions and consistent coding standards improves
Reusability	5	Modular components and reusable functions reduce code duplication and simplify maintenance.
Documentation / Comments	4	Important modules and complex logic are documented with comments.
Overall Score	4.5	The project follows good coding practices with a modular, maintainable, and reusable architecture.